2025 Second International Conference on Pioneering Developments in Computer Science & Digital Technologies (IC2SDT) | 979-8-3315-7725-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/IC2SDT68218.2025.11383595

Enhanced DINOv2-Based Framework for Multi-Class Gastrointestinal Abnormality Classification in Capsule Endoscopy

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***Abstract*—Millions of individuals all over the globe are affected by the gastrointestinal (GI) disorders, which require early diagnosis so that they can be treated appropriately. Video capsule endoscopy has grown to be a excellent diagnostic tool with the ability to give clear images of the digestive system. Nevertheless, large volumes of frames generated during a single examination, can be a significant burden on the manual review by clinicians. To address this issue, we presented an automated classification system, which is based on deep learning, and uses DINOv2 architecture. The model is trained to label ten GI conditions Angioectasia, Bleeding, Erosion, Erythema, Foreign Body, Lymphangiectasia, Normal, Polyp, Ulcer and Worms with labeled VCE images. The system achieved an average accuracy of 93.31% percent, on the test data, and it shows good results on various measures, such as precision, recall, F1-score, AUC-ROC, specificity and sensitivity. Unlike previous studies by Avala et al. [13], which optimized a DINOv2 ViT-S/14 encoder on the Capsule Vision training, our model freezes the ViT-B/14 encoder and trains a small classifier, boosting accuracy but reducing training expenses. The system will enhance the computer-aided GI diagnostics by allowing the accelerated and more dependable judgement of the endoscopic images.**

***Index Terms*—Gastrointestinal Abnormalities, Video Capsule Endoscopy (VCE), Deep Learning, Self-Supervised Learning, DINOv2, Vision Transformers, Medical Image Classification, Transformer-based Models.**

I. Introduction

Gastrointestinal (GI) diseases are a major health problem in the world, with high morbidity and mortality rates being reported by Handa et al. [1]. Timely and proper diagnosis plays a very crucial role in treatment and better outcomes of the patients. Video Capsule Endoscopy (VCE) is an endoscopic technique offering a minimally invasive method of visualizing the entire GI tract. Nonetheless, tens of thousands of frames are produced in each procedure, and manual verification is time-consuming, prone to errors, and clinicians [1].

The development of artificial intelligence (AI) and deep learning, in particular, has demonstrated a good chance to overcome these difficulties, as it allows detecting and classifying GI abnormalities automatically. Although standard supervised models are promising approaches, their use is limited by the limited availability of labeled medical data and patient and imaging condition variability.

Recently, self-supervised learning has become a potent paradigm that can be used to overcome these limitations. DINOv2, a Vision Transformer architecture trained on scale unlabeled images has shown the best performance in extracting transferable and generalizable features to a variety of computer vision tasks, discussed by Oquab et al. [3]. Its use in medical imaging, organ segmentation, and disease detection further underscores its possible use in the analysis of GI disease [4].

In this paper, we utilize and improve DINOv2 to classify GI diseases based on VCE. In particular, we propose architectural enhancements and task-tailored optimisations that enhance classification results compared to baseline implementations and reduce exposure to large annotated datasets. Our works indicate the possibility of implementing complex self-supervised frameworks that can effectively, precisely, and in a large scale, apply AI-assisted diagnosis to gastrointestinal endoscopy.

This research uses DINOv2 to come up with an AI-based

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framework to classify GI abnormalities in VCE images. With the model trained on a dataset that consists of ten different GI conditions, we can test the model with regards to its accuracy in identifying and classifying these abnormalities. We aim to evaluate the possibility to use self-supervised learning models like DINOv2 to enhance the effectiveness and accuracy of GI diagnosis

II. Related Work

Xinrui Song et al. [5] proposes a training-free deformable image registration framework named DINO-Reg which makes use of off-the-shelf DINOv2 features. They only require simple fine-tuning to apply their method, which involves inputting the pre-trained encoder to learn semantically rich descriptors on medical scans and a discrete optimizer, which computes deformation fields, to obtain competitive registration accuracy on benchmark datasets. Arnav Samal et al. [6] introduce an ensemble with convolutional neural networks (CNNs) and vision transformers to detect the presence of abnormalities in video capsule endoscopy (VCE) frames. Their method on more than 50,000 labeled images with ten classes achieves a balanced accuracy of 86.34% and mean AUC-ROC of 0.9908, demonstrating the usefulness of hybrid architectures in the capture of both local and global information. M. F. Ahamed et al. [7] create a deep learning pipeline with an EfficientNetB5 architecture, enhanced with customized data-augmentation and regularization techniques in GI tract disease diagnosis. This model is tested on several conditions, including ulcerative colitis and polyps with the test accuracy of 98.89%, which indicates the efficiency of transfer learning and high-level augmentation in the classification of endoscopic images. D. Mukhtorov et al. [8] discuss model interpretability, which they achieved through the combination of Grad-CAM and a ResNet152 backbone to classify endoscopic images in the KVASIR dataset. Their explainable framework does not only give a 93.46 % accuracy on validation but also gives the visual heat maps to indicate the areas of discrimination, which earns AI-driven diagnosis more credibility as a clinician. Lee and Park [9] explore self-supervised pre-training on the DINO architecture on restricted labeled data (110 per cent of the whole) on endoscopy, dermoscopy, and bone marrow microscopy tasks. They demonstrate that DINO pre-training followed by light supervised fine-tuning surpasses fully supervised baselines, validating self-supervised learning as a key strategy for low-data medical settings. Smith et al. [10] present a two-phase transfer learning framework for GI disease detection and segmentation using models such as Xception, InceptionResNetV2, and VGG16. Their extensive evaluation on public endoscopic datasets reveals that InceptionResNetV2 fine-tuned in two stages delivers consistent, generalized performance, highlighting the utility of structured transfer learning pipelines. Huang et al. [11] compile the GastroNet-5M dataset of over five million GI endoscopic images and compare self-supervised methods (SimCLRv2, MoCov2, DINO) against supervised ImageNet pre-training. Their experiments on downstream GI tasks demonstrate that in-domain self-supervised pre-training, particularly with DINO, consistently outperforms natural image pre-training, emphasizing the importance of domain-specific representation learning.

While prior works have explored CNN–ViT ensembles [6], transfer learning with EfficientNet [7], interpretability frameworks [8], and self-supervised pre-training on medical data [9] [11], our approach is distinct in directly adapting the latest DINOv2 large-scale self-supervised vision transformers for end-to-end abnormality classification in video capsule endoscopy (VCE) images. Unlike training-free registration frameworks that rely solely on feature extraction [5], or domain-specific pre-training pipelines that require additional in-domain pretext tasks [9] [11], we leverage the strong generalization of DINO encoders as frozen or partially fine-tuned backbones, coupled with a lightweight classification head. This strategy enables us to achieve state-of-the-art classification accuracy (93.31%) across ten VCE abnormality classes, outperforming ensemble and transfer learning baselines, while also maintaining efficiency by minimizing architectural complexity and additional training overhead. According to Avala et al. [13], the research was a proposal of a DINOv2-based deep learning system to automatically categorize gastrointestinal abnormalities in video capsule endoscopy (VCE) images. Their approach relied on the DINOv2 ViT-S/14 backbone that is fine-tuned on the Capsule Vision 2024 dataset and is further enhanced with extensive data augmentation and a multi-layer classification head to obtain strong results on all ten abnormality classes. We base our work on this and employ a fixed feature extractor in the form of a frozen DINOv2 ViT-B/14 encoder, eliminating the necessity to fine-tune the model to its full extent. We also optimize pre-processing by using simplified resizing and center-cropping schemes, re-architecture the classifier head to reduce the complexities of the architecture, and use optimized batch sizes and modern hardware to accelerate training. These optimizations lead to a computationally-efficient pipeline without sacrificing classification accuracy, and show that architecture-specific and training changes could drive future progress in transformer-based analysis of VCE imagery. The comparison of our study and a reference study on 10-class VCE abnormality classification shown in Table I.

Recent advances in self-supervised and transfer learning have demonstrated the potential of large vision models, particularly DINO/DINOv2, in medical imaging tasks. Training-free registration using DINOv2 features has achieved competitive results [5]. Hybrid CNN–ViT ensembles have been explored specifically for VCE abnormality classification [6]. EfficientNet-based pipelines with tailored augmentation have reported high accuracy on GI conditions [7]. Explainability with Grad-CAM improves clinician trust while maintaining strong performance [8]. Systematic reviews also highlight that self-supervised pre-training with DINO-style objectives is effective in low-label medical settings [9], while comprehensive surveys on medical ViTs discuss design choices and transfer strategies [10]. In GI endoscopy specifically, foundation models and pre-training strategy choices materially affect downstream performance [11].

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TABLE I
COMPARISON OF PROPOSED AND REFERENCE METHODS ON 10-CLASS VCE (CAPSULE VISION 2024) [13].

Aspect	Proposed Approach	Existing Approach [13]
Problem & classes	10-class VCE abnormality classification (Angioectasia, Bleeding, Erosion, Erythema, Foreign Body, Lymphangiectasia, Normal, Polyp, Ulcer, Worms).	Same 10 classes.
Dataset provenance	Capsule Vision 2024 dataset; "50k+" labeled frames.	Capsule Vision 2024; explicit source breakdown (SEE-AI, KID, Kvasir-Capsule, AIIMS).
Split	Uses 80/20 stratified split for train/test. Test total reported 16,132 (matches challenge validation size).	Uses official split 37,607 train / 16,132 validation, with per-source counts.
Preprocess & aug	Resize $\rightarrow$ random crop to 224×224, horizontal flip p=0.5, ImageNet normalization; val/test: resize + center-crop.	Resize to 224×224; random flips, rotations, random resized crop, brightness tweaks; val: resize + center-crop; ImageNet stats.
Backbone	DINOv2 ViT-B/14 encoder used as frozen feature extractor.	DINOv2 ViT-S/14 loaded and fine-tuned on medical data.
Classifier head	384 $\rightarrow$ 128 $\rightarrow$ 10 (ReLU then softmax).	384 $\rightarrow$ 256 $\rightarrow$ 10 (two FC layers).
Frozen vs fine-tune	DINOv2 encoder frozen; only head trained.	DINOv2 backbone fine-tuned.
Optimizer & LR	Adam, LR = 1e$-$6 (very low), cross-entropy.	Adam, LR = 1e$-$3 with scheduler; cross-entropy.
Batch & epochs	Batch 64, 10 epochs.	Batch 128, 6 epochs (early stopping considered).
Hardware	NVIDIA RTX A6000.	NVIDIA RTX 3080.

Unlike training-free frameworks that use only frozen features [5], or in-domain pretext pipelines [11], our approach adapts a DINOv2 encoder as a frozen or lightly fine-tuned backbone with a compact classifier head for end-to-end VCE abnormality classification, attaining high accuracy with minimal architectural complexity.

III. PROPOSED APPROACH

The overall pipeline shown in Fig. 1 follows a self-distillation framework built on Vision Transformers (ViTs).

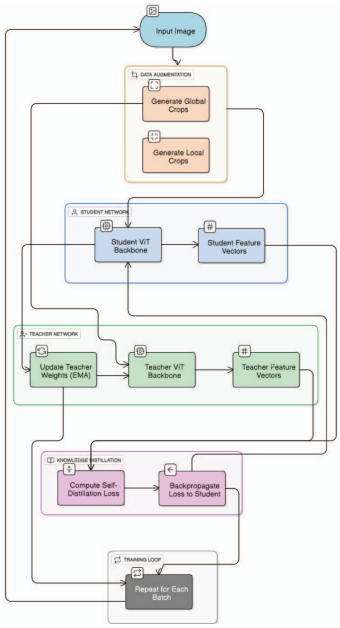


Fig. 1. DINOv2-based VCE image classification pipeline.

The process unfolds in the following stages:

1) **Data Augmentation:** Each input image undergoes strong augmentation by generating both global and local crops, ensuring multi-scale contextual information.
2) **Student Network:** The augmented views are passed through the Student ViT backbone, producing Student Feature Vectors.
3) **Teacher Network:** A Teacher ViT backbone, whose weights are updated using Exponential Moving Average (EMA) of the student, processes the same augmented views to generate Teacher Feature Vectors.
4) **Knowledge Distillation:** A self-distillation loss is computed between the Student and Teacher Feature Vectors. The error is then backpropagated to update the Student network.
5) **Training Loop:** Steps 1–4 are repeated iteratively for each training batch until model convergence.

Through this design, the proposed approach enables the student model to learn rich and generalizable representations without requiring explicit labels, making it effective for downstream classification tasks.

A. Dataset Description

The dataset comprises over 50,000 labeled VCE images spanning 10 classes: Angioectasia, Bleeding, Erosion, Erythema, Foreign Body, Lymphangiectasia, Normal, Polyp, Ulcer, and Worms. The distribution of training samples across these classes is illustrated in Fig. 2.

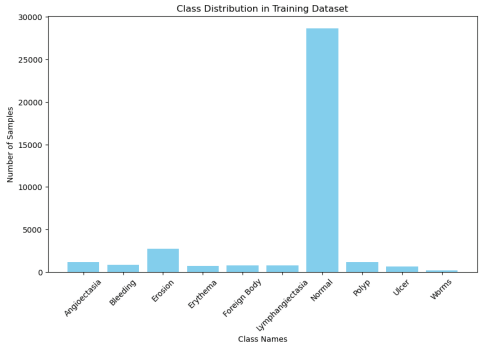


Fig. 2. Dataset distribution.

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The dataset is organized in a class-folder structure compatible with `torchvision.datasets.ImageFolder`. It was obtained from the Capsule Vision 2024 Challenge [12]. Representative samples from the Angioectasia, Bleeding, Lymphangiectasia, and Normal classes are presented in Fig. 3.

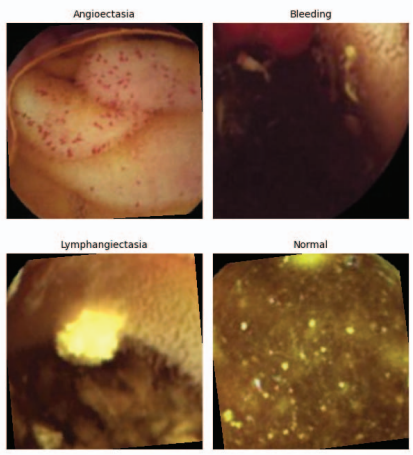


Fig. 3. Sample training images [12].

B. Data Preparation and Preprocessing

The dataset is split into training and testing sets using an 80/20 ratio with stratification to preserve class distributions.

For training, the following data augmentation and preprocessing steps are applied: resizing to 256 pixels, random cropping to 224 pixels, horizontal flipping with probability $p = 0.5$, and normalization using ImageNet statistics. For testing, images are resized and center-cropped to 224 pixels, followed by the same normalization.

Feature extraction is performed using a DINOv2 ViT-B/14 encoder with an embedding dimension of 384. A lightweight classifier head first reduces the feature dimension from 384 to 128 using a ReLU activation, and then maps to 10 classes via a linear layer, with softmax applied during inference.

C. Training Configuration

The proposed model was trained using the Adam optimizer with a learning rate of 1×10^{-6} and cross-entropy loss. Training was conducted with a batch size of 64 for 10 epochs on an NVIDIA RTX A6000 GPU. To ensure reproducibility, random seeds were fixed, and the best-performing model state was saved based on validation accuracy.

Figures 4 and 5 illustrate the training and validation performance across epochs. The accuracy curve (Fig. 4) shows stable improvement without divergence between training and validation, indicating that the model generalizes well. ext: On the same note, the loss curve (Fig. 5) is also convergent, which also proves the lack of overfitting.

IV. Results

This section describes the results of the experiment that was carried out with the help of the given approach. We state a summary as well as a class-wise results on the Capsule Vision 2024 dataset, in conjunction with comparisons with existing work. Also, we examine learning. represent behavior in accuracy and loss curves and measure computational efficiency in. parameters, footprint, and energy consumption term.

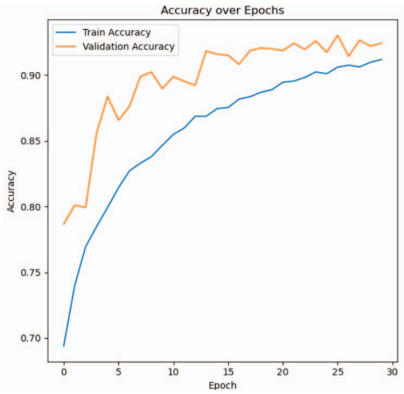


Fig. 4. Training and validation accuracy over epochs.

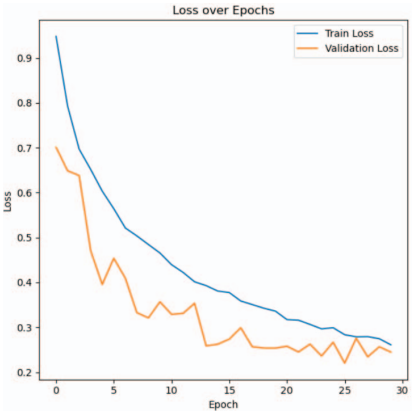


Fig. 5. Training and validation loss over epochs.

A. Confusion Matrix and ROC/AUC

An overall accuracy of the model is 93.31% per-class precision, recall and F1-scores are strong. The normalized confusion matrix (Fig. 6) is mostly diagonal meaning that the classification by category is reliable with most of the mistakes being made between classes that are visually similar (e.g., *erythema* vs. *erosion*). The one-vs-rest ROC analysis once again supports the presence of strong discrimination, with a mean AUC of 0.9917.

B. Comparison with Existing Method

In order to put the effectiveness of our approach in a context we compare the optimized frozen. Lightweight classifier of DINOv2 ViT-B/14 vs. the fine-tuned DINOv2 ViT-S/14 baseline presented by Avala et al. [13]. Table II summarizes per-class and aggregate metrics, highlighting consistent improvements in accuracy, weighted F1, and robustness on minority classes (e.g., *worms*, *erythema*). In addition, we report model efficiency indicators such as parameter count,

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TABLE II
EXPANDED COMPARISON OF REPORTED METRICS AND ARTIFACTS FOR 10-CLASS VCE CLASSIFICATION.

Aspect	Proposed Approach	Existing Approach [13]
Evaluation metrics reported	Accuracy; per-class Precision/Recall/F1; macro & weighted averages; confusion matrix; AUC-ROC, AP, sensitivity, specificity.	Accuracy; per-class Precision/Recall/F1; macro/weighted averages; confusion matrix; Avg AUC, specificity, sensitivity summarized.
Headline results	Accuracy 93.31%; weighted Precision/Recall/F1 ≈ 0.937/0.933/0.933; totals 16,132 images.	Accuracy 91%; Macro-F1 ≈ 0.71, Weighted-F1 ≈ 0.91.
Per-class highlights (examples)	Normal F1 0.9754; Worms F1 0.9927; Erythema F1 0.6448; Polyp F1 0.6821.	Normal F1 0.97; Worms F1 0.40 (low recall); Erythema F1 0.35; Polyp F1 0.58.
AUC / specificity / sensitivity	Reports ROC/AUC evaluation (mean AUC-ROC listed among metrics).	Summarizes Avg AUC 0.9914, Specificity 0.9847, Sensitivity 0.7849.
Baselines / comps	Compares multiple CNNs without aug/fine-tune (e.g., VGG19, ResNet50V2, MobileNetV2).	Compares against challenge baselines/teams with ranking table.
Compute footprint	Reports 22.06M params, ∼84,192 KB memory, 1,680 s compute, ∼61,000 J energy.	Not reported. (No corresponding section.)
Curves & plots	Accuracy/Loss curves; normalized confusion matrix; per-class table.	Training/validation graphs; confusion matrices; per-class table; leaderboard table.

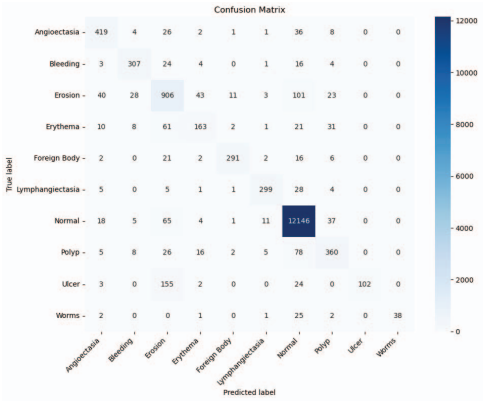


Fig. 6. Confusion matrix.

memory usage, training time, and energy consumption, which were not addressed in the baseline work.

C. Class-wise Performance

Table III presents the detailed classification report.

TABLE III
CLASSIFICATION REPORT ON THE TEST SET.

Class	Prec.	Rec.	F1	Support
Angioectasia	0.8803	0.7988	0.8376	497
Bleeding	0.8466	0.8914	0.8684	359
Erosion	0.7207	0.7732	0.7460	1155
Erythema	0.6023	0.6936	0.6448	297
Foreign Body	0.8404	0.9294	0.8827	340
Lymphangiectasia	0.8661	0.8863	0.8761	343
Normal	0.9876	0.9635	0.9754	12287
Polyp	0.5979	0.7940	0.6821	500
Ulcer	0.9673	0.9331	0.9483	286
Worms	0.9855	1.0000	0.9927	68
Overall Acc.		**0.9331**		16132
Macro Avg	0.8295	0.8660	0.8454	16132
Weighted Avg	0.9368	0.9331	0.9327	16132
#Params		22.06M		
Memory (KB)		84192		
Energy (J)		61000		
Time (s)		1680		

The proposed model achieves an overall accuracy of 93.31%. Macro precision, recall, and F1-score are 0.8295, 0.8660, and 0.8454, respectively, while weighted averages are consistently higher (≈0.93). The highest F1-scores are observed for *worms* (0.9927), *normal* (0.9754), and *ulcer* (0.9483). By contrast, relatively lower scores occur for *erythema* (0.6448) and *polyp* (0.6821), which are visually challenging and often confused with other inflammatory or protruding lesions.

V. CONCLUSION

This work introduced an enhanced DINOv2-based framework for automated multi-class classification of gastrointestinal abnormalities in video capsule endoscopy (VCE) images. By leveraging a frozen ViT-B/14 encoder and a lightweight classifier head, our approach achieves a competitive accuracy of 93.31% across ten clinically relevant classes, with consistently high precision, recall, F1-scores, and AUC-ROC values. Compared to existing DINOv2 fine-tuned baselines, the proposed pipeline not only improves overall accuracy but also demonstrates superior robustness on minority classes such as *worms* and *erythema*, while maintaining reduced training cost, memory footprint, and energy consumption. The results highlight the promise of self-supervised vision transformers as powerful and generalizable vision backbones to medical imaging, especially in low-label image settings. Following practice, these systems may facilitate quicker and more trustworthy interpretation of endoscopic images, easing the pressure of clinicians and allowing them to diagnose GI conditions promptly. This framework will be expanded to time series analysis of full-length VCE videos, incorporation of explainability modules to simulate clinical trust and adaptation to larger gastrointestinal data sets to enhance generalizability in the future.

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